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From plasma proteomics Mendelian randomization to neuropathological validation: The potential role of CTSH-GRN-TMEM106B and TREM2-IL-34 networks of microglia in Alzheimer's disease

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A R T I C L E I N F O

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A B S T R A C T

Background: Alzheimer's disease (AD) has complex pathophysiological features, and the interaction between peripheral and central systems has recently been considered to contribute to its occurrence and development. The complexity of the disease may not be fully captured by currently available exploration methods, such as biomarkers, which are limited by correlation associations. Comprehensive analytical methods are necessary to elucidate the association networks between different tissues, thereby further understanding the potential mechanisms of this disease.
Methods: This study used a bidirectional Mendelian randomization (MR) framework to establish causal relationships between peripheral biomarkers and AD. It examined 734 plasmas and 151 cerebrospinal fluid (CSF) proteins as exposure factors, compared to AD diagnosis (causal relationship from peripheral to AD). In reverse, the study also explored 1296 brain protein outcomes using an AD genetic risk tool (causal relationship from AD to brain pathology). Additionally, this study integrated single-cell transcriptome data from prefrontal cortex specimens and validated findings in AD postmortem human brain tissue through immunohistochemistry and immunofluorescence, followed by protein-protein interaction (PPI) analysis using the STRING database to construct networks.
Results: The bidirectional MR analysis revealed different proteomic characteristics, including elevated levels of cathepsin H (CTSH), signal regulatory protein alpha (SIRPA), and transmembrane protein 106B (TMEM106B) in plasma, and increased bone sialoprotein (BSP), interleukin-34 (IL-34), and immunoglobulin-like transcript 2 (ILT-2) in CSF, all exhibiting a positive correlation with AD risk. In contrast, higher plasma granulin (GRN), triggering receptor expressed on myeloid cells 2 (TREM2), sialic acid-binding immunoglobulin-like lectin-9 (SIGLEC9), and CSF SIGLEC7 exhibited protective effects. Reverse MR analysis revealed that AD is associated with the upregulation of CTSH, SIRPA, TMEM106B, IL-34, GRN, TREM2, SIGLEC9, and SIGLEC7 expression in the brain. Neuropathological validation confirmed the cortical overexpression of CTSH, GRN, TMEM106B, SIGLEC9, and TREM2 in AD specimens, consistent with MR analysis. According to the PPI network analysis, the regulatory networks of CTSH-GRN-TMEM106B and TREM2-IL-34 may be closely linked to AD.
Conclusions: This study established a connection between peripheral biomarkers and the central system, validated in AD postmortem human brain tissue samples, revealing a microglial CTSH-GRN-TMEM106B and TREM2-IL-34-related networks that may be involved in AD occurrence and development.

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1. Introduction

Alzheimer's disease (AD) is a major neurodegenerative disorder affecting the elderly and presents an increasingly severe global health challenge, clinically characterized by the gradual deterioration of memory and cognitive functions (Estévez-Silva et al., 2022). The pathological changes of AD include the formation of plaques through extracellular amyloid β-protein (Aβ) deposition and neurofibrillary tangles resulting from excessive phosphorylation of intracellular tau protein ("Global, regional, and national incidence, prevalence, and years lived with disability for 328 diseases and injuries for 195 countries, 1990–2016: a systematic analysis for the Global Burden of Disease Study, 2016," 2017; Gu et al., 2022; Kaiser, 2018). Existing treatment methods for AD, including cholinesterase inhibitors and NMDA antagonists, only provide symptomatic relief and do not change disease progression (Ma and Gang, 2004; Marucci et al., 2021; Reisberg et al., 2003). The recent approval of controversial anti-amyloid biologics, such as aducanumab and lecanemab, highlights the importance of new therapeutic targets (Budd Haeberlein et al., 2022; Terao and Kodama, 2024). New targets for AD treatment should address the complex, multifaceted nature of the disease beyond the traditional amyloid cascade hypothesis and target various potential factors that contribute to the disease.

Emerging evidence has recently indicated that the interaction between peripheral circulation and the central system plays a significant role in the early pathogenesis of AD (Park et al., 2019). Neuropathological data from human AD studies revealed that the loss of cortical tight junction proteins is associated with the pathology of AD as the disease progresses (Yamazaki et al., 2019). The impairment of the endothelial barrier may interact with amyloid and tau pathology in cases of blood-brain barrier (BBB) damage. This interaction may lead to the infiltration of neurotoxic plasma proteins, thereby disrupting brain homeostasis and exacerbating the condition further. This phenomenon emphasizes the valuable potential of blood and cerebrospinal fluid (CSF) proteome in developing new therapeutic approaches. However, there are certain limitations in this area of study. First, there is a spatial inconsistency between circulating biomarkers and brain pathology. For instance, plasma Aβ42 levels decrease as brain amyloid accumulation occurs (Smirnov et al., 2022; H. Y. Wang et al., 2020). Second, large-scale omics approaches often lack cellular-level analysis, which can obscure the roles played by different cell types. Third, there is a temporal gap between relevant changes in plasma and CSF circulation and postmortem pathological validation.

In this context, we initiated a comprehensive study aimed at exploring the dynamic interactions between peripheral circulation, the pathology of the central nervous system (CNS), and the mechanisms underlying the onset of AD.

2. Materials and methods

2.1. Bidirectional Mendelian randomization (MR) study design

This study was designed using a bidirectional MR framework. Forward MR aims to explore the causal effect of peripheral proteins on the pathogenesis of AD, where protein levels in plasma and CSF serve as exposure factors and AD risk as the outcome. Reverse MR focuses on analyzing the impact of AD on brain proteins, where the exposure factor is the risk of AD, and the result is the protein levels in the cortical brain after death.

2.2. Data sources and instrumental variable screening

2.2.1. Positive MR analysis

2.2.1.1. Exposure data: Plasma and CSF protein quantitative trait loci (pQTLs). The Nature Genetics consortium contributed data for plasma pQTLs (Zheng et al., 2020), combining results from five genome-wide association studies (GWASs) to identify 3606 pQTLs linked to 2656 proteins (Folkersen et al., 2017; Suhre et al., 2017; Sun et al., 2018; Yao et al., 2018). Additionally, CSF proteomic data were sourced from research published in Nature Neuroscience (Yang et al., 2021), which documented 274 pQTLs associated with 184 proteins. This study is based on a large-scale cohort that integrates plasma data from over 14, 000 individuals who have undergone a standardized quality control process, thereby enhancing its statistical power. These pQTLs were selected based on the following criteria: Cis-acting variants to demonstrate genome-wide significance ($p < 5 \times 10^{-8}$), the exclusion of the major histocompatibility complex region located on chromosome 6 (26–34 Mb), and linkage disequilibrium clumping with a threshold of $r^2 < 0.001$. Lastly, the final analysis included 738 cis-pQTLs for 734 plasma proteins (Supplemental Table 1) and 154 cis-pQTLs for 151 CSF proteins (Supplemental Table 2).

2.2.1.2. Outcome data: Patients with AD. Summary statistics for the GWAS were derived from the European Alzheimer's and Dementia Biobank consortium, comprising 111,326 clinically diagnosed cases of AD or their 'proxy' counterparts, alongside 677,663 control subjects (Bellenguez et al., 2022).

2.2.2. Reverse MR analysis

2.2.2.1. Exposure data: Patients with AD. The data collected from the European Alzheimer's and Dementia Biobank consortium included 111,326 cases of AD and 677,663 control samples, as mentioned above (Bellenguez et al., 2022). This analysis identified 21 independent single-nucleotide polymorphisms (SNPs) with a significance threshold of $p < 5 \times 10^{-8}$ and an r^2 value of less than 0.001.

2.2.2.2. Outcome data: Brain proteomics. Cortex protein GWAS data were sourced from the National Institute on Aging Genetics of AD and included 1296 proteins (Greenfest-Allen et al., 2024). This data quantified postmortem protein levels using tandem mass tag mass spectrometry across 458 brain samples. The link to download this data is given in Supplemental Table 3.

2.3. MR analysis process

In the forward MR analysis, the inverse variance weighted (IVW) method was used to combine the effects of multiple tools, and the Wald ratio estimation was employed in cases involving a single tool (Deng et al., 2022). All analyses were conducted using the TwoSampleMR package (version 0.5.7) in R software (version 4.2.3) (S. Wang et al., 2023). A Bonferroni correction was used to set the genome-wide significance threshold and ensure statistical validity, resulting in a threshold of $\alpha = 0.05$ divided by 885 tested proteins, which equates to 5.65×10^{-5}.

In the reverse MR analysis, IVW was used as the primary method of analysis. Sensitivity analyses, including Cochran's Q (retaining results with $p > 0.05$), MR-Egger intercept (also retaining $p > 0.05$), and leave-one-out validation tests, were performed, requiring that the results from all analyses maintain over 80 % consistency. Significance across the proteome range was established through false discovery rate (FDR) correction by adjusting for the expected proportion of false positives among significant results at $p < 0.05$ (Verbanck et al., 2018).

Instead of biomarker identification, this reverse MR design

Table 1
Detailed information on AD cases and the control group.

Case	Group	Age	Gender	Cause of death	Underlying disease
1	AD	77	Male	Hemorrhage within the plaque of the left anterior descending coronary artery.	Severe coronary heart disease, myocardial infarction.
2	AD	80	Male	Acute onset of severe coronary heart disease.	Severe coronary heart disease
3	AD	65	Female	Acute hemorrhagic shock caused by multiple sharp force injuries to the neck.	None
4	Control	71	Female	Acute onset of severe coronary heart disease.	Severe coronary heart disease
5	Control	74	Male	Small cell lung cancer with multi-organ metastasis and systemic multi-organ failure.	Small cell lung cancer
6	Control	84	Male	Acute onset of severe coronary heart disease.	Coronary heart disease, chronic granulomatous sclerosing nephritis.

strategically explores the proteomic disturbances caused by, focusing on pathologically relevant downstream effects that can be assessed postmortem. This panel, comprising 1296 proteins and encompassing 85 % of the druggable human proteome, identifies potential therapeutic targets.

2.4. Single-cell transcriptome analysis

2.4.1. Sample source and selection criteria

The single-cell sequencing data used in this study were sourced from a publicly available dataset published by Lau et al. (2020) (Lau et al., 2020). This original dataset includes 21 postmortem tissue samples from the prefrontal cortex, specifically from Brodmann areas 6, 8, and 9, all sourced from the South West Dementia Brain Bank. Additionally, 12 samples were from patients with AD, and nine were from normal control individuals. The RNA-sequencing (RNA-seq) analysis of AD brain tissue in their study identified 14,386 genes, covering 5684 cells.

2.4.2. Dimensionality and clustering

Dimensionality reduction and clustering analyses were performed using the R package 'Seurat' (Butler et al., 2018). The filtered dataset was first processed using the Log-Normalize and ScaleData algorithms. Subsequently, the FindVariableFeature function was applied to identify genes for additional principal component analysis, and the percentage variation between successive principal components was calculated. If this percentage dropped below 5 %, 10 principal components were selected for the subsequent application of the FindNeighbors function. The R package 'Harmony' was used to correct for batch effects between samples. The analysis incorporated the uniform manifold approximation and projection (UMAP) technique. Cluster analysis was conducted using the FindClusters function. Characteristic of markers for specific immune cell types were identified by leveraging the ScType database, facilitating the assignment of putative cell types to each population of cells (Ianevski et al., 2022).

Subsequently, the expression levels of genes encoding target proteins were assessed across various cell types. A differential expression analysis was conducted using the Wilcoxon Rank Sum test to evaluate the expression of identified protein-coding genes implicated in AD within specific brain cell types, comparing their expression to other cell types. Genes exhibiting an average Log2 fold change greater than 0.5 and an FDR-adjusted p-value lower than 0.05 were categorized as enriched within the respective cell type. The results of the dimensionality reduction were visualized using the R package 'plot1 cell' (Wu et al., 2022), while additional results were depicted using the R packages 'Seurat' and 'ggplot2'.

2.5. Pathological verification of AD specimens

2.5.1. Ethical compliance and sample acquisition

Postmortem human brain specimens were acquired from the Chinese Brain Bank Center, Department of Forensic Medicine, Huazhong University of Science and Technology, Wuhan, China. Autopsy samples with a history of AD before death were selected, along with age-matched autopsy samples as controls, as depicted in Table 1. All brain tissues (temporal lobe cortex and hippocampus) were obtained from body donors, ensuring no risk to living individuals. Each specimen was assigned a de-identified numerical code, and all personal identification information (names and ID numbers) was permanently anonymized to maintain donor privacy. Written informed consent was obtained from the legal representatives of the donors, and the research protocol was approved by the Ethics Committee of Huazhong University of Science and Technology. The relevant ethical materials are provided in the appendix (ethical approval number: 2024 [S201]).

2.5.2. Tissue processing and immunohistochemistry (IHC)

Paraffin-embedded specimens from the temporal cortex and hippocampal regions were sliced into 4 μm sections, baked to improve adhesion, and then stored at −20 °C until further analysis. The sections were dewaxed in xylene for IHC and then rehydrated using a series of graded ethanol solutions. Antigen retrieval was conducted in a microwave using a specified buffer, such as citrate buffer at pH 6.0, to facilitate the exposure of target antigens. The sections were treated with a 3 % hydrogen peroxide solution for 10 min to block endogenous peroxidase activity. The slides were then incubated with primary antibodies in a humidified chamber at 4 °C overnight to ensure optimal binding. After three washes with Tris-buffered saline, horseradish peroxidase-conjugated secondary antibodies were applied and incubated for 1 h at room temperature. Signal amplification was achieved using 3, 3′-diaminobenzidine chromogen for a specified duration under controlled conditions. Finally, the sections were counterstained with hematoxylin for 3 min, dehydrated using graded ethanol, cleaned in xylene, and then mounted with a resinous medium to preserve the samples for further examination. All antibodies used are presented in Appendix 1.

2.5.3. Immunofluorescence (IF) staining

For IF staining, sections were blocked with 5 % donkey serum in 0.3 % Triton X-100/PBS for 1 h, then incubated with primary antibodies against the target proteins at 4 °C for 48 h. After PBS washes, sections were stained with Alexa Fluor-conjugated secondary antibodies (1:500, Invitrogen) for 2 h at room temperature. Nuclei were counterstained with DAPI (Beyotime, C1005), and slides were mounted with ProLong Gold antifade reagent (Thermo Fisher). Images were acquired using a confocal microscope (Zeiss LSM 900) with a 20 × objective. All antibodies used are presented in Appendix 1.

2.5.4. Imaging and statistical analyses

Stained sections were scanned uniformly with a NanoZoomer S360 digital slide scanner (Hamamatsu Photonics K.K., Japan) under constant bright-field illumination. The optical density measurements for the target markers were quantified within designated regions of interest using ImageJ software (National Institutes of Health, Bethesda, MD, USA). To minimize discrepancies among different brain samples during autopsy collection, we selected areas for observation and statistical analysis where the quantity and density of neurons and glial cells were relatively uniform.

Statistical analyses were conducted using the GraphPad Prism software (version 9; GraphPad Software, San Diego, CA). Independent samples t-tests were used to compare differences between groups, and p

Table 2
MR results for plasma and CSF proteins markedly linked to AD after Bonferroni-adjusted.

Tissue	Protein	UniProt ID	SNP [a]	Effect allele	OR (95 % CI) [b]	*P*-value	PVE	F statistics	Author
Plasma	*CTSH*	P09668	rs34593439	A	1.06 (1.03, 1.08)	6.12e-06	24.98 %	1098.94	Sun
Plasma	*GRN*	P28799	rs5848	T	0.79 (0.74, 0.84)	2.19e-12	2.85 %	96.68	Sun
Plasma	*SIRPA*	P78324	rs6136377	G	1.03 (1.02, 1.05)	1.37e-05	65.48 %	6260.64	Sun
Plasma	*TMEM106B*	Q9NUM4	rs10950398	A	1.16 (1.09, 1.23)	1.92e-06	3.57 %	118.37	Emilsson
Plasma	*TREM2*	Q9NZC2	rs114812713	G	0.67 (0.58, 0.78)	6.95e-08	1.23 %	39.89	Emilsson
Plasma	*SIGLEC9*	Q9Y336	rs2075803	G	1.03 (1.02, 1.04)	4.30e-05	75.05 %	9926.96	Sun
CSF	*BSP*	P21815	rs6855426	G	1.33 (1.17, 1.51)	9.34e-06	3.74 %	32.46	Yang
CSF	*IL-34*	Q6ZMJ4	rs36097154	G	2.13 (1.51, 3.01)	1.85e-05	16.80 %	168.58	Yang
CSF	*ILT-2*	Q8NHL6	rs2114511	G	1.48 (1.25, 1.76)	4.04e-06	21.58 %	229.77	Yang
CSF	Siglec-7	Q9Y286	rs2075803	G	0.42 (0.28, 0.64)	4.30e-05	4.44 %	38.79	Yang

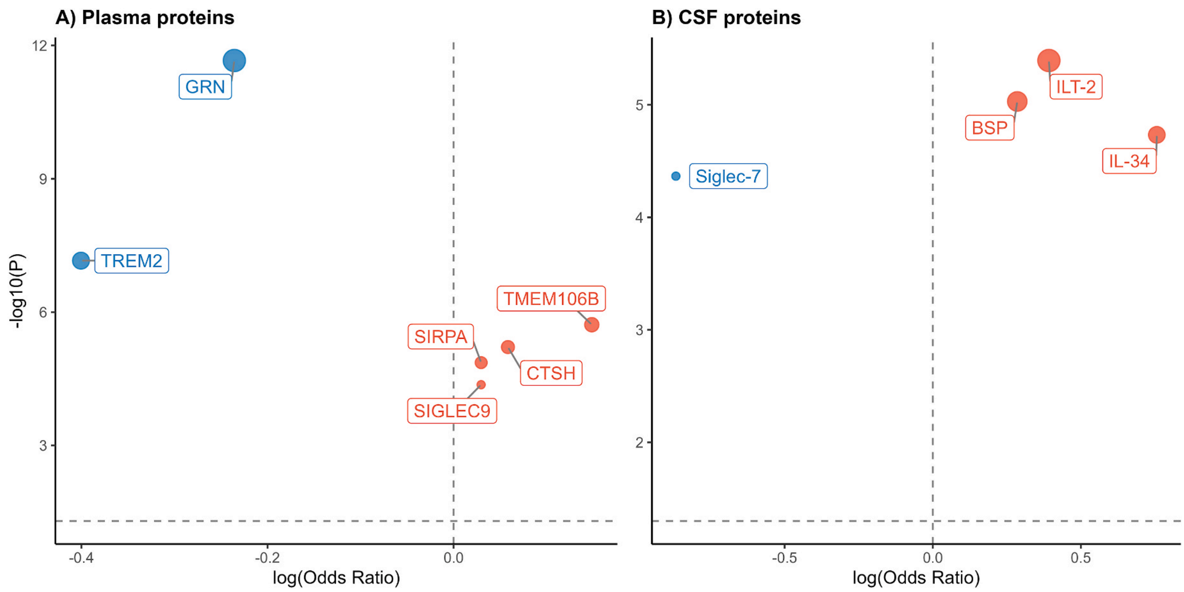

Fig. 1. A volcano plot was constructed to identify 10 proteins as prospective targets in AD from a comprehensive dataset of 734 plasma proteins (A) and 151 CSF proteins (B). This analysis utilized the Wald ratio or the IVW approach. Here, CSF refers to cerebrospinal fluid, and PVE denotes the proportion of variance explained. Notably, some results are derived from our previously published research (S. Wang et al., 2025).

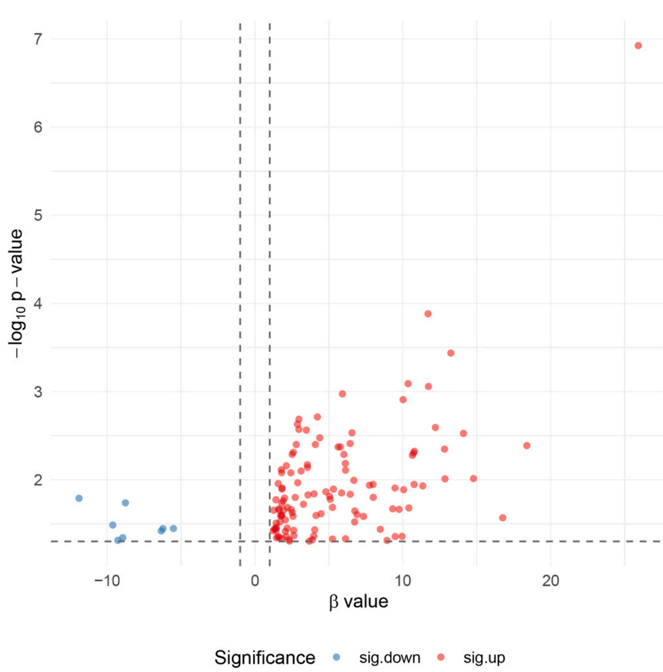

Fig. 2. The 125 proteins associated with AD in the brain.

< 0.05 was considered statistically significant. Data are presented as mean ± standard error of the mean (AD group n = 3, control group n = 3).

2.6. Protein-protein interaction (PPI) analysis

The PPI network was constructed by entering all important proteins (including cathepsin H (CTSH), plasma granulin (GRN), and triggering receptor expressed on myeloid cells 2 (TREM2), among others) into STRING (version 12.0) with default parameters (medium confidence = 0.4, full interactome mode) (Szklarczyk et al., 2015). All first-order interactions were retained regardless of node centrality, with edge weights determined by a composite scoring algorithm of STRING integrating seven evidence channels (coexpression, experiments, databases, among others).

3. Results

3.1. MR analysis identifies causal roles of proteins in AD

This study systematically evaluated the causal associations between 734 plasma proteins and 154 CSF proteins with AD using two-sample MR analysis with a Bonferroni-corrected threshold of $p < 5.65 \times 10^{-5}$. A total of 10 proteins were found to be significantly associated with the risk of developing AD for the first time. These proteins included CTSH, GRN, signal regulatory protein alpha (SIRPA), transmembrane protein

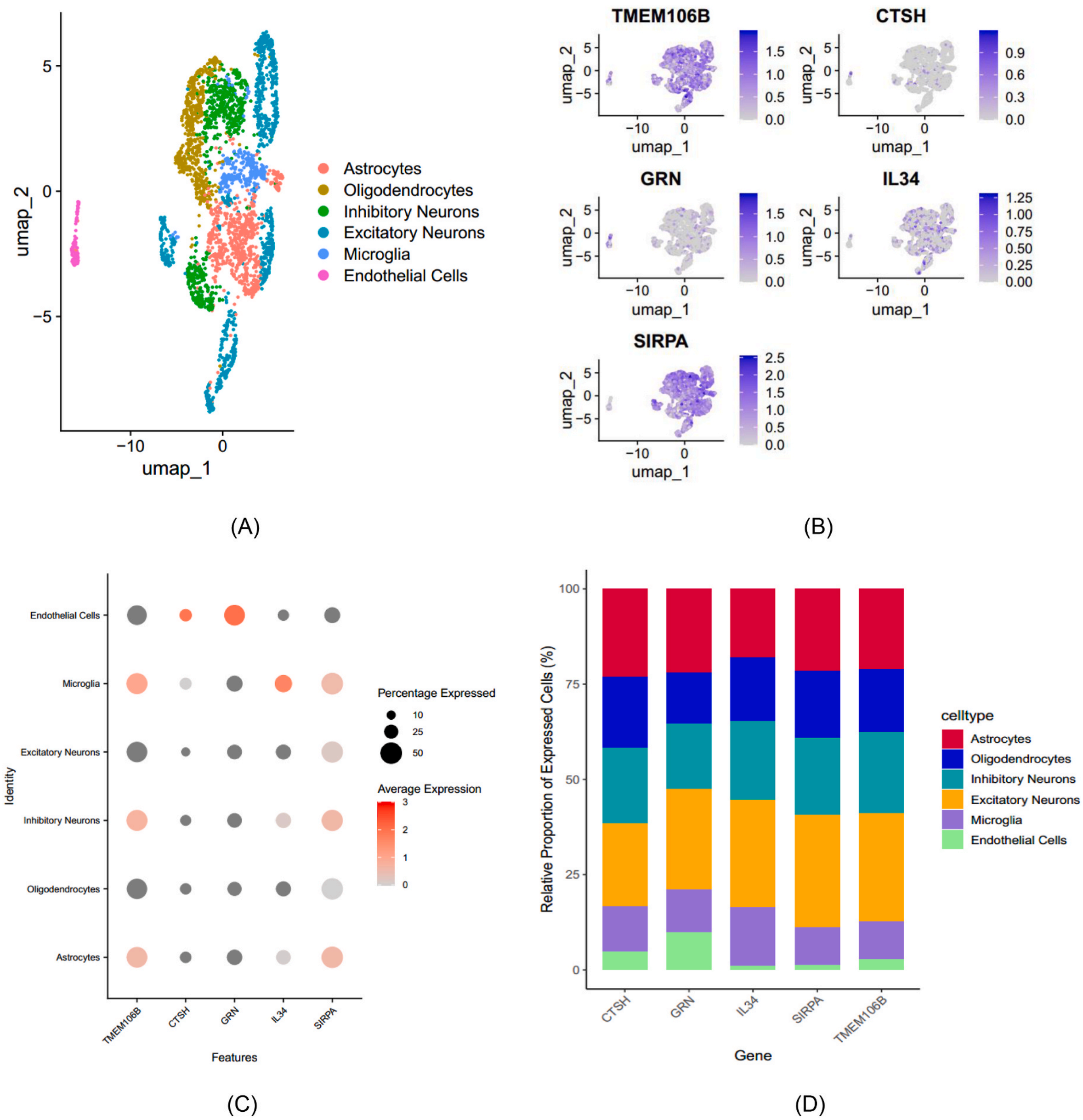

Fig. 3. Single-cell transcriptomic profiling of causal protein-coding genes in brains of patients with AD. (A) UMAP visualization of cellular clusters. (B) Detection of MR-identified protein-coding genes. (C) Cell-type-specific expression patterns. Percentage expressed: The percentage of cells within a specific cell cluster that exhibit detectable expression of a given gene. Average expression: The mean expression level of a gene calculated across all cells within a specific cell cluster, including those where the gene was not detected. (D) Spatial mapping of causal gene signatures. Relative proportion of expressed cells: The percentage of cells belonging to a specific cell cluster relative to the total number of high-quality, transcriptionally active cells analyzed in the entire dataset or defined sample group.

106B (TMEM106B), TREM2, sialic acid-binding immunoglobulin-like lectins (SIGLEC)-9/-7, bone sialoprotein (BSP), interleukin-34 (IL-34), and immunoglobulin-like transcript 2 (ILT-2) (Table 2 and Fig. 1).

The key findings included that the risk of AD was significantly increased by elevated levels of plasma CTSH (odds ratio [OR] = 1.06, $p = 6.12 \times 10^{-6}$), SIRPA (OR = 1.03, $p = 1.37 \times 10^{-5}$), TMEM106B (OR = 1.16, $p = 1.92 \times 10^{-6}$), CSF BSP (OR = 1.33, $p = 9.34 \times 10^{-6}$), IL-34 (OR = 2.13, $p = 1.85 \times 10^{-5}$), and ILT-2 (OR = 1.33, $p = 9.34 \times 10^{-6}$). Conversely, increased levels of plasma GRN (OR = 0.79, $p = 2.19 \times 10^{-12}$), TREM2 (OR = 0.67, $p = 6.95 \times 10^{-8}$), SIGLEC9 (OR = 0.67, $p = 6.95 \times 10^{-8}$), and CSF SIGLEC7 (OR = 0.42, $p = 4.30 \times 10^{-5}$) were associated with reduced risk of AD.

3.2. Supplementary bidirectional MR analysis

The reverse MR analysis identified 125 cortical proteins with an FDR <0.05 causally linked to AD. Among these proteins, 116 were found to be upregulated, while nine were downregulated. The prioritized proteins converged on three functional axes: Myeloid homeostasis (for instance, TREM2 and IL-34), lysosomal processing (including CTSH and GRN), and intercellular communication (represented by SIRPA and SIGLEC-family receptors). Furthermore, 7 of the 125 proteins associated with AD in the brain (Fig. 2) exhibited significant overlap with proteins linked to the risk of AD identified in the forward MR analysis derived from blood and CSF. These overlapping proteins included neurodegeneration-related proteins, such as TMEM106B and GRN, as well as immunomodulatory proteins, including TREM2 and SIRPA. Additionally, the proteolytic system components CTSH and sialic acid-

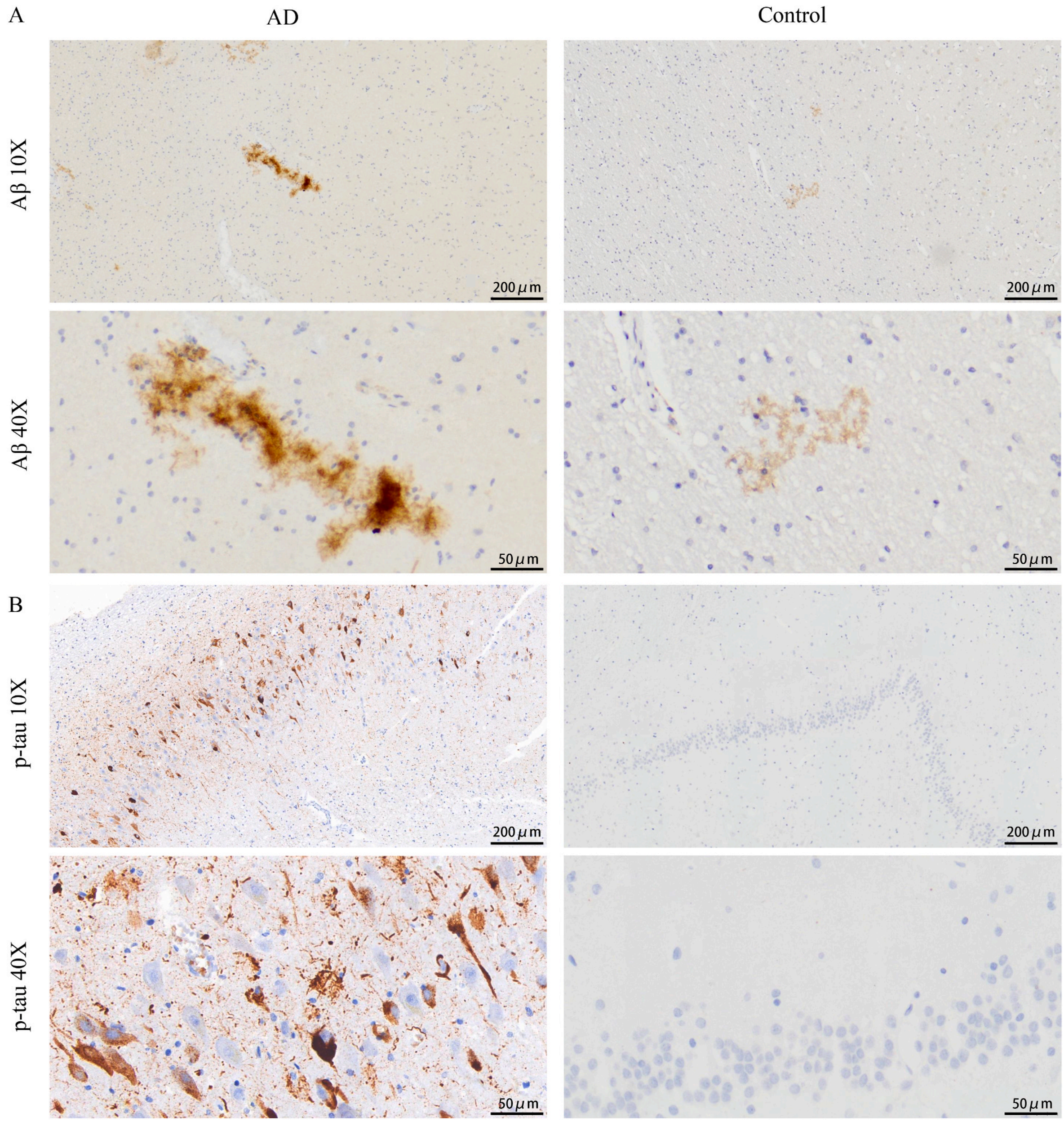

Fig. 4. (A) Aβ and (B) p-tau pathology in autopsy samples.

binding proteins, like SIGLEC-9, were identified. The validity of all significant associations was confirmed using MR-Egger regression, with the intercept test yielding a *p*-value greater than 0.05, as well as the weighted median method. Cochran's Q test also indicated no significant heterogeneity ($p > 0.05$), and leave-one-out analysis demonstrated no bias driven by any single SNP.

3.3. Single-cell validation of differential proteins

Moreover, single-cell RNA sequencing (scRNA-seq) was used to investigate gene expression levels in the brain tissue of patients with AD. The scRNA-seq data were subjected to quality control. Six clusters were identified in the AD patient group and categorized as astrocytes, endothelial cells, inhibitory neurons, excitatory neurons, microglia, and oligodendrocytes (Fig. 3A). Notably, the expression of genes linked with the causal proteins was validated independently in several cell types. The expression of 5 out of 8 protein-coding genes was detectable in the brain tissue of patients with AD, whereas the expressions of *TREM2*, *SIGLEC-9*, and *SIGLEC-7* were undetected (Fig. 3B). Fig. 3C illustrates the single-cell expression of these coding genes in every cluster. Cell type-specific expression profiling demonstrated TMEM106B enrichment in microglia, excitatory neurons, and astrocytes. CTSH exhibited a highly specific upregulation in endothelial cells but was transcriptionally suppressed across neuronal lineages. The average expression level of GRN is the highest in endothelial cells. IL-34 exhibited the highest average expression level within the microglial cluster. SIRPA expression was highest in astrocytes, albeit with a relatively modest increase compared to some other cell types, while being markedly suppressed in endothelial cells. Analysis of undetected key genes revealed that TREM2, SIGLEC-9, and SIGLEC-7 exhibited expression levels close to the detection threshold, potentially due to probe sensitivity limitations or the dominance of post-transcriptional regulation mechanisms. scRNA-seq analysis revealed distinct expression patterns of causal protein-coding genes specific to different cell types in the brains of patients with AD, as depicted in Fig. 3D.

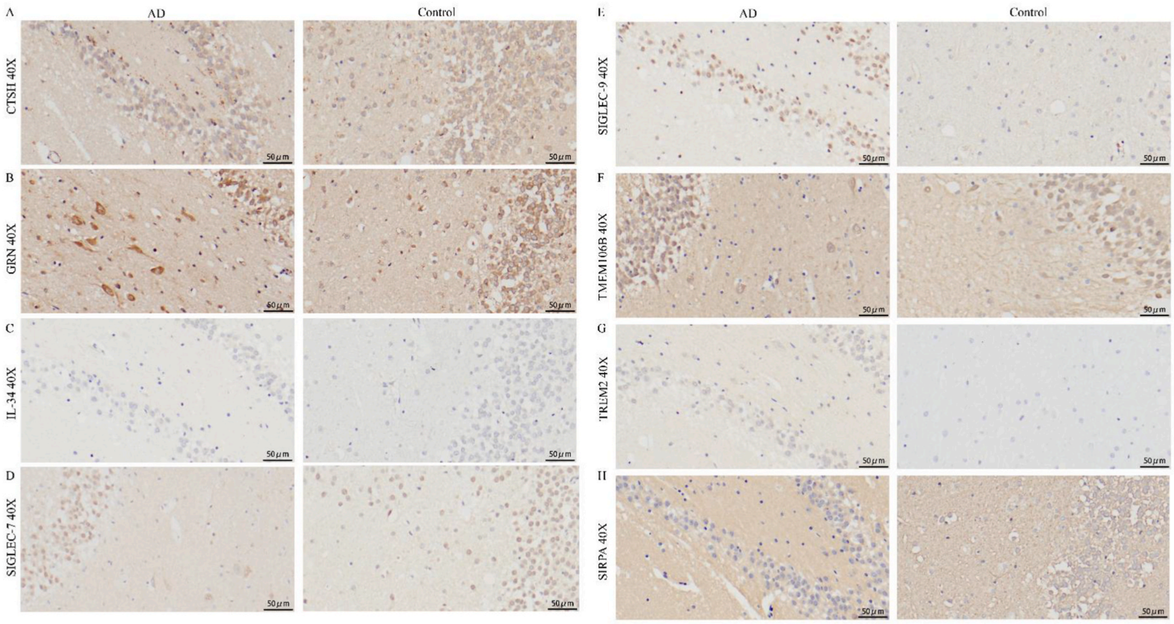

Fig. 5. IHC staining of biomarkers in the hippocampus region of autopsy samples.

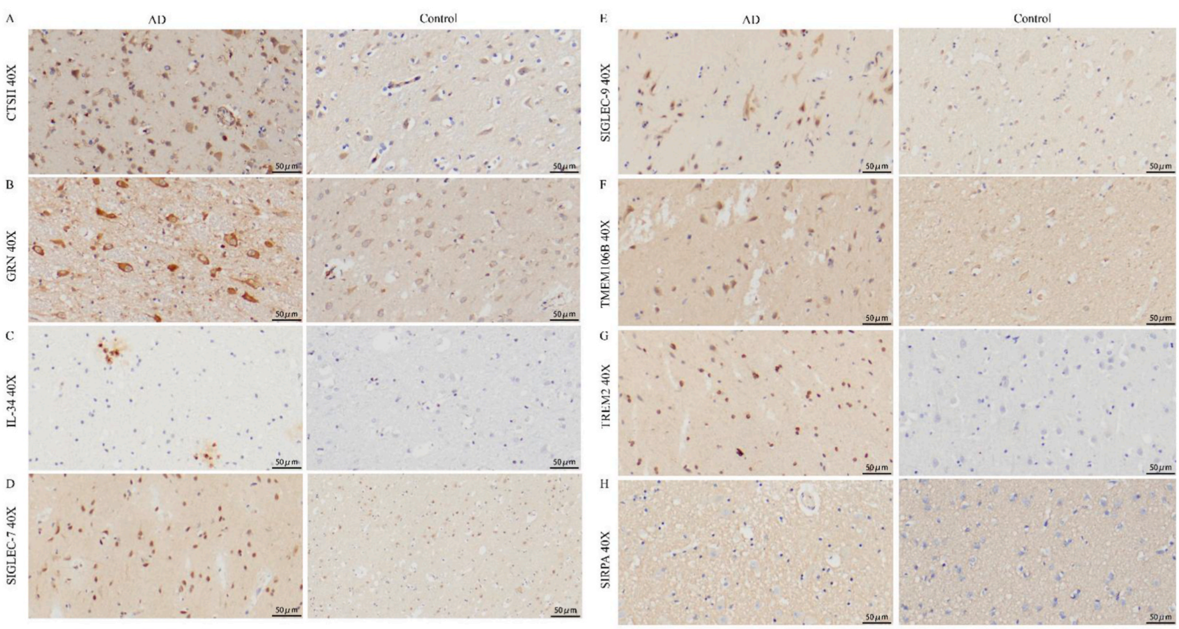

Fig. 6. Biomarker IHC staining of the temporal cortex in autopsy samples.

3.4. Pathological validation of AD samples

Immunohistochemical analysis of Aβ plaque deposition and phosphorylated tau (p-tau) neurofibrillary tangles confirmed the distinct pathological features of AD and non-AD postmortem samples. In AD cases, Aβ peptides, generated from amyloid precursor protein through β- and γ-secretase cleavage, form dense extracellular plaques (Fig. 4A), while highly p-tau aggregates manifest as intracellular neurofibrillary tangles in the hippocampus region (Fig. 4B). This result aligns with the pathological characteristics of AD, where Aβ plaques lead to neuronal damage, and p-tau-induced microtubule disassembly promotes synaptic dysfunction and apoptosis. In contrast, non-AD brains exhibited sparse Aβ immunoreactivity, without plaque formation, and minimal p-tau staining in the hippocampus region.

3.5. Immunohistochemical staining of differential proteins in autopsy samples

GRN expression was significantly increased in postmortem samples from the hippocampus region of the AD group compared to the control group. In contrast, the differences in other markers between the two groups did not exhibit statistical significance (Figs. 5 and 7A). Additionally, five markers displayed statistically significant upregulation in the temporal cortex of both groups, including CTSH ($p = 0.032$), GRN ($p = 0.006$), SIGLEC-9 ($p = 0.045$), TREM2 ($p = 0.009$), and TMEM106B ($p = 0.045$). Although SIGLEC-7 and IL-34 also exhibited an upward trend, they did not reach statistical significance ($p = 0.231$ and 0.234, respectively) (Figs. 6 and 7B). SIRPA did not reveal any significant positive staining in any of the samples.

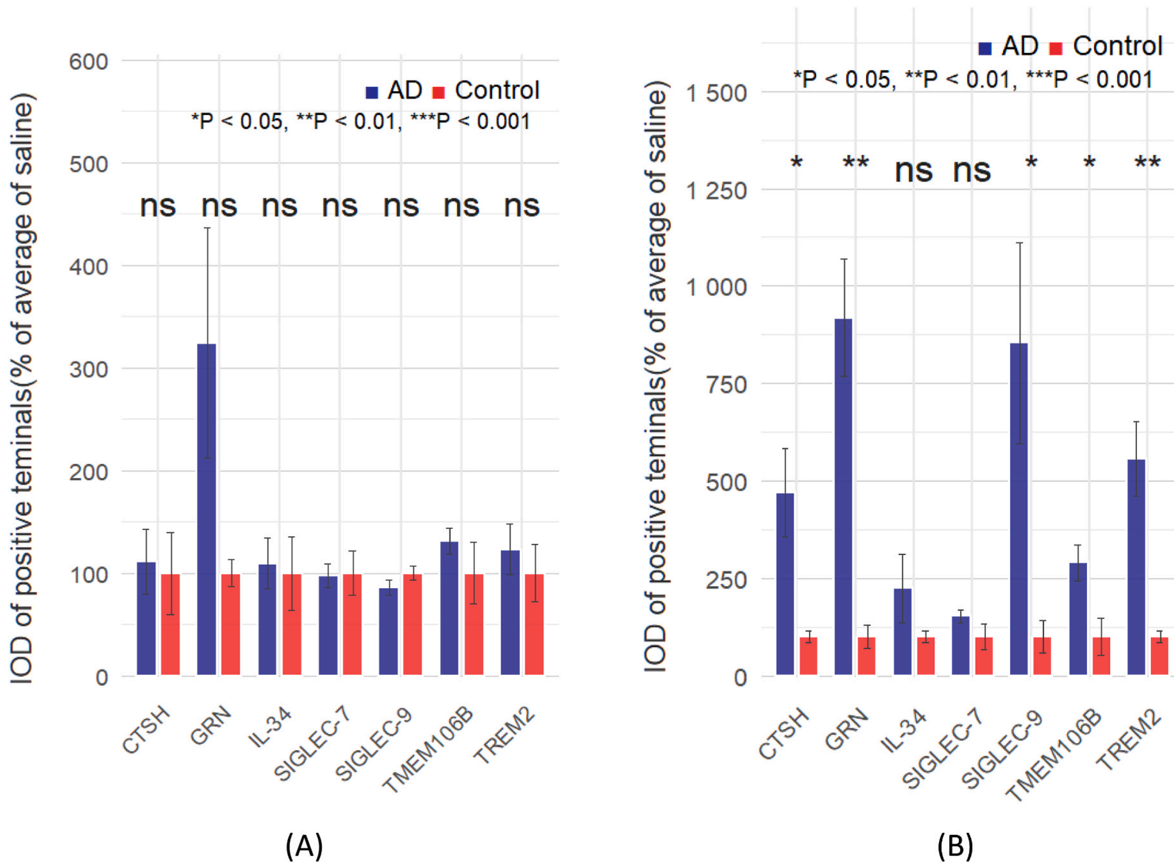

Fig. 7. Semi-quantitative results of biomarkers in autopsy samples: (A) hippocampus and (B) cortex.

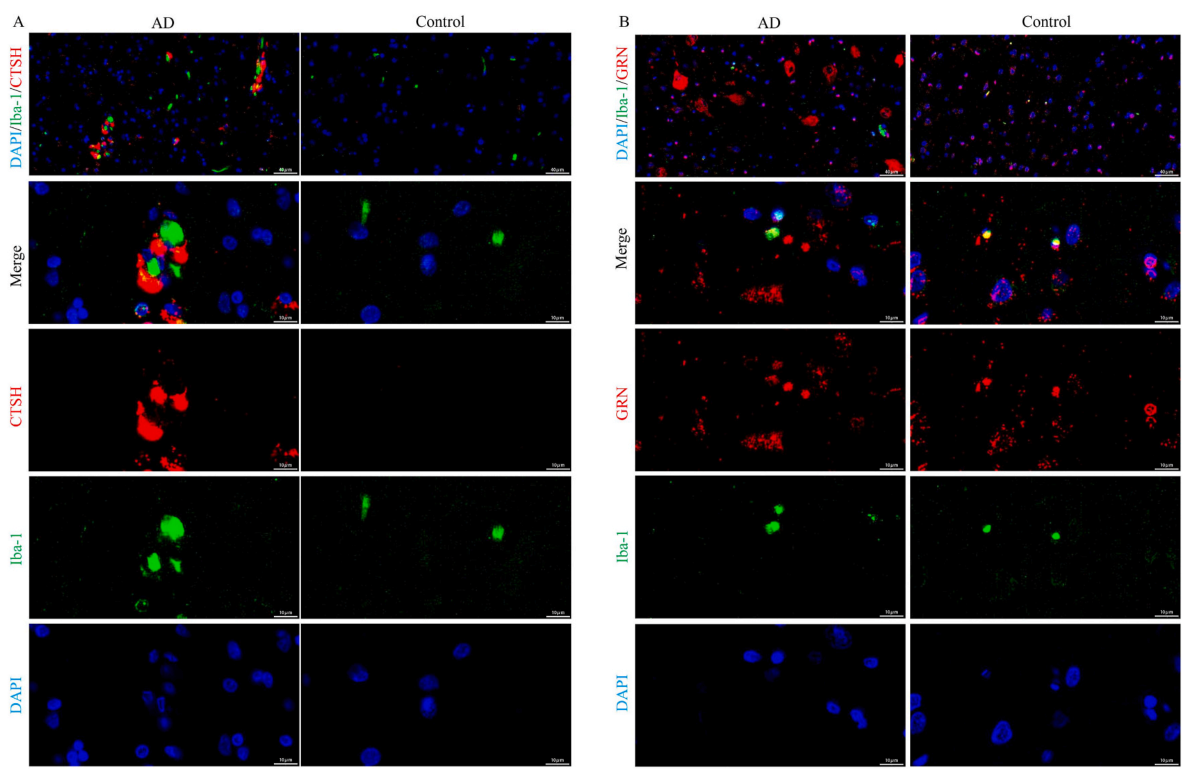

Fig. 8. Microglial co-localization patterns of neurodegeneration-associated proteins in the cortex of AD and control groups. (A) DAPI/Iba-1/CTSH. (B) DAPI/Iba-1/GRN methods: Nuclei (DAPI, blue), microglia (anti-Iba1, green), and target proteins (specific antibodies, red). Scale bars are located in the bottom right corner of the image.

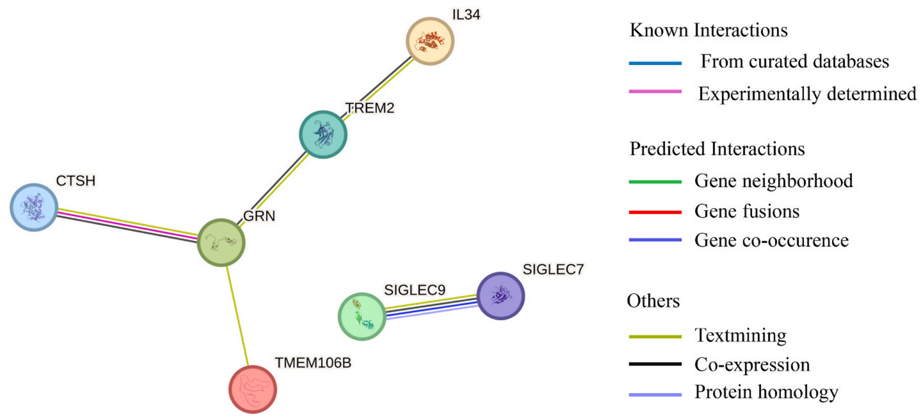

Fig. 9. PPI network centered on microglia.

Table 3
Hierarchical functional scores and statistical significance of the microglial PPI network.

Level	Function	Protein 1	Protein 2	Combined score
1	Core lysosomal regulatory factors	GRN	TREM2	0.553
1	Core lysosomal regulatory factors	GRN	TMEM106B	0.906
2	Immune signaling axis	IL34	TREM2	0.685
2	Immune signaling axis	SIGLEC7	SIGLEC9	0.762
3	Protease-chaperone pair	CTSH	GRN	0.492

3.6. IF staining of differential proteins in autopsy samples

Based on the results of immunohistochemical staining, dual IF staining on CTSH, GRN, and microglia was performed in autopsy samples. The results demonstrated significant co-localization of CTSH-positive areas with microglia in the AD group (Fig. 8A), while positive staining was rarely observed in the control group. GRN exhibited positive expression, primarily co-localizing with microglia in the control group. In contrast, it co-localized with microglia in the AD group and displayed widespread expression in other cells and the extracellular space (Fig. 8B).

3.7. Analysis of PPI networks

STRING analysis generated an interaction network centered on microglia, which was divided into three functional tiers (Fig. 9, Table 3). The first tier focused on core lysosomal regulatory factors, where the interaction between GRN and TMEM106B was prominent, with the highest combined score of 0.906, primarily supported by database annotations. The connectivity of GRN-TREM2 also exhibited a moderate level at this point, with a score of 0.553, supported by coexpression and text mining evidence. The second tier emphasized the immune signaling axis, where the interaction between IL-34 and TREM2 scored 0.685, mainly driven by coexpression patterns and database organization. Besides, SIGLEC7 and SIGLEC9 demonstrated high sequence homology (score: 0.967), indicating an evolutionary relationship between these genes. The third tier involved a protease-chaperone pair, particularly the interaction between CTSH and GRN, which scored 0.492. This interaction was supported by experimental validation (score = 0.301) and transcriptomic correlation.

4. Discussion

This study used a comprehensive four-tiered verification framework to explore the dynamic interactions between peripheral circulation, pathology of the CNS, and the mechanisms underlying the onset of AD. First, MR was applied to 734 plasma and 151 CSF proteins to identify causative biomarkers associated with AD, avoiding bias from reverse causation in observational studies. Second, reverse MR analysis of brain proteins confirmed the CNS relevance of peripheral circulating biomarkers. Third, existing single-cell transcriptomic data from the prefrontal cortex demonstrated the cellular tendency of candidate target analysis. Lastly, neuropathological validation in AD postmortem specimens confirmed the relevance to disease stages. Through this approach, we aim to explore CTSH-GRN-TMEM106B and TREM2-IL-34 networks as key microglial drivers in AD progression. TREM2 is a microglia-specific receptor regulating phagocytosis and inflammation (Yoo et al., 2023). GRN and TMEM106B form a lysosomal module modulating microglial homeostasis (Takahashi and Strittmatter, 2025). SIGLEC7 and SIGLEC9 are immunomodulatory siglecs predominantly expressed on microglia in the brains of patients with AD (Linnartz-Gerlach et al., 2014). Collectively, these findings position microglial lysosomal-immune crosstalk as a core mechanistic node, offering new therapeutic entry points for AD modulation. The network provides new directions for understanding the mechanisms behind AD.

CTSH—a lysosomal cysteine protease family member—is crucial for protein degradation and antigen presentation (M. Wang et al., 2022). Studies have found altered CTSH expression in the temporal cortex of patients with late-onset AD (Chen et al., 2020). GRN, which encodes granulin precursor, is a crucial chaperone protein for maintaining lysosomal homeostasis (Logan et al., 2021). The protein encoded by the GRN gene plays a key role in the growth, survival, function, and maintenance of neurons and microglia in the mammalian CNS (Rhinn et al., 2022). Mutations in the GRN gene are often associated with frontotemporal dementia (FTD), and numerous studies have suggested that these mutations may promote AD pathogenesis (Ingannato et al., 2023). MR results of this study revealed that increased levels of CTSH in plasma are associated with an increased risk of AD (OR = 1.06). Moreover, IHC results demonstrated increased positive expression of CTSH in postmortem temporal cortex samples from patients with AD, indicating that peripheral circulating CTSH can serve as an early diagnostic marker. Recent studies have identified that peripheral CTSH can activate amyloid precursor protein through protease activity (Lü et al., 2007). CTSH is specifically localized in the microglia of patients with AD as demonstrated by the findings of IF experiments. It was hypothesized that the central pathological role of this protein is highly dependent on the activation of microglia. As central immune monitors, the lysosomal system of microglia is overactivated in AD, leading to CTSH-mediated protease overload (Y. S. Wang et al., 2023) and subsequently driving neuroinflammation and abnormal processing of amyloid precursor protein. In contrast, increased levels of GRN in plasma are associated with a decreased risk of AD (OR = 0.79), and reverse MR revealed elevated GRN in the prefrontal cortex of patients with AD. Meanwhile, IF results revealed that in the autopsy samples of the control group, GRN was mainly localized in microglia cells. However, in the AD group, there was widespread ectopic expression of GRN. This provides cytological evidence in support of the hypothesis of BBB damage. Microglia cells maintain neuroprotective functions through GRN under normal physiological conditions; however, when the BBB permeability is impaired, vascular-derived GRN abnormally infiltrates the brain parenchyma. Its accumulation outside microglia cells may form toxic aggregates, promoting the transformation of the protective mechanism to a pathological state. scRNA-seq analysis indicated that the protective effect associated with plasma GRN may originate from or be modulated by its predominant expression in brain endothelial cells. We hypothesize that the compromised permeability of the BBB may contribute to a compensatory increase in GRN levels in brain tissue, thereby providing a protective effect. However, dysregulation or accumulation of GRN may exacerbate AD pathogenesis.

The genetic polymorphism of TMEM106B is associated with the progression of various neurodegenerative diseases (Brady et al., 2013) and serves as a key regulatory factor that bridges lysosomal homeostasis (Arrant et al., 2018) and the pathogenesis of neurodegenerative diseases. It has drawn widespread attention in AD-related studies in recent years (Jiao et al., 2023). It is an essential transmembrane protein that regulates lysosomal acidification and affects the clearance efficiency of protein aggregates by maintaining lysosomal pH balance (Feng et al., 2021). MR results of this study indicated that elevated plasma levels of TMEM106B were associated with an increased risk of AD (OR = 1.16). This association was supported by brain protein MR results and is consistent with findings from autopsy samples of AD pathology. scRNA-seq results identified that TMEM106B was significantly enriched in microglia, excitatory neurons, and astrocytes. Similar findings have also been observed in autopsies of other neurodegenerative diseases, including AD (Brady et al., 2013), amyotrophic lateral sclerosis, and FTD (Chang et al., 2022; Fan et al., 2022; Jiang et al., 2022; Schweighauser et al., 2022). However, the therapeutic development of TMEM106B-related targets has yielded contradictory results (Edwards et al., 2024; Perez-Canamas et al., 2021), which must be interpreted cautiously. Although antibody-based therapies targeting TMEM106B have demonstrated targeted involvement in preclinical models, their ineffectiveness against the accumulation of tau suggests their effects may vary according to the disease stage (Labadorf et al., 2023). Future studies should focus on a few key areas to address these complexities: (1) the temporal changes of TMEM106B isoforms in disease progression; (2) the characterization of interactions between neurons and microglia mediated by lysosomal protein homeostasis; (3) the development of specific conformational regulators rather than relying on general neutralization strategies.

TREM2, a microglia-specific receptor, provides neuroprotective functions by mediating the phagocytosis of Aβ and regulating inflammatory responses (Qin et al., 2021). Elevated levels of TREM2 are associated with the recruitment of microglia to amyloid plaques (Cuyvers and Sleegers, 2016). The results of this study revealed that the increased levels of TREM2 in plasma are associated with a reduced risk of AD (OR = 0.67); however, reverse MR and pathological validation of autopsy samples demonstrated increased levels in the cerebral cortex, indicating an urgent need to identify the optimal activation window for TREM2.

IL-34 is a newly identified cytokine that acts as an alternative ligand for the colony-stimulating factor 1 receptor, working in conjunction with macrophage colony-stimulating factor (Lin et al., 2008; Zuroff et al., 2020). Previous studies have revealed that IL-34 inhibits the differentiation of monocytes into macrophages, reducing their phagocytic capacity for abnormal Aβ (Zuroff et al., 2020). MR results of this study indicated that increased levels of IL-34 in CSF are associated with an increased risk of AD (OR = 2.13). scRNA-seq data analysis from this study demonstrates that IL-34 is explicitly expressed in microglia. This microglia-derived IL-34 may act in an autocrine or paracrine manner to potentially enhance microglial signaling (Alzoubi et al., 2023), which could contribute to the phagocytosis of abnormal Aβ.

SIGLEC, as a glycoimmune checkpoint receptor, inhibits the activation of immune cells when interacting with specific sialic acid glycan ligands (Smith et al., 2023). The SIGLEC family embodies the complex interplay between glycoimmune regulation and neurodegenerative pathology (Lünemann et al., 2021). The MR analysis of this study confirmed the neuroprotective potential of SIGLEC9 (plasma OR = 0.67, 95 % confidence interval [CI]: 0.58–0.78) and SIGLEC7 (CSF OR = 0.42, 95 % CI: 0.28–0.64); in line with their established roles as glycoimmune checkpoint regulators (Smith et al., 2023). The data analysis from scRNA-seq did not detect signals for SIGLEC7 and SIGLEC9. The elevation of SIGLEC7 levels in CSF demonstrated a protective effect against AD (OR = 0.42). The SIGLEC9 protein was significantly accumulated in AD autopsy cortical samples. The observed lack of transcriptional activity in scRNA-seq profiles may be attributed to the primary factors, such as the regulation occurring after the translation of functional isoforms that bind sialoglycans, which may be undetectable through conventional transcriptional profiling methods (Hansson et al., 2023). The initial finding of protein-level dysregulation in AD cortical lesions corresponded with the recognized function of SIGLEC9 as a modulator of glycoimmune checkpoints, which can be explained through several biologically plausible mechanisms. First, changes in sialylation associated with Aβ may result in post-translational stabilization, which aligns with the typical ligand-binding properties of SIGLEC9. Second, there may be a dynamic redistribution of SIGLEC9 proteins toward the membranes of microglia in the context of AD, resulting in observable immunohistochemical signals.

There may be two possible explanations for significant differences in molecular expression between the hippocampus and cortical regions during the validation of AD autopsy samples. First, the microvascular density of the cerebral cortex is much higher than that of the hippocampal region (Bell and Ball, 1981). It is more likely to come into contact with circulating factors through the permeable BBB sites (such as meningeal lymphatic vessels). This allows peripheral proteins to enter in large quantities, which may potentially exacerbate the cortical lesions of AD. Second, the cortical layers are directly connected to the subarachnoid cavity, enabling the signal molecules transmitted through CSF to penetrate more easily than in the deeply embedded hippocampus. This may accelerate the protein homeostasis dysfunction of the cortex. Hippocampus degeneration is a key trigger for AD; however, the vulnerability of the cerebral cortex to environmental imbalances may further accelerate the progression of the disease. This observation highlights a dual pathogenic mechanism that necessitates targeted treatment strategies focused on specific areas of the brain.

The network structure of microglial dysfunction was outlined by analyzing the overall PPI architecture. The lysosomal interaction complex of GRN-TMEM106B ranked highest in this PPI network with a composite score of 0.906, consistent with the findings of this study. The IL-34-TREM2 axis scored 0.685, serving as a spatial encoder in the network, where the gradient of neuronal IL-34 guides the polarization of microglia expressing TREM2 toward the interface of Aβ and tau proteins. This finding aligns with the results of the current study, which demonstrated that CSF IL-34 increases the risk of AD and that TREM2 levels are elevated in the cortices of patients with AD. The CTSH-GRN axis scored 0.492, indicating interactions between proteases and chaperones driving pathological changes. Experimental evidence scored 0.301, demonstrating that CTSH mediates the cleavage of GRN, consistent with the findings of this study, revealing increased AD risk associated with plasma CTSH and GRN accumulation in astrocytes. This suggests that excessive activity of peripheral CTSH may produce GRN fragments capable of crossing the BBB and initiating cortical aggregation. This interaction may demonstrate a positive feedback loop, including the loss of GRN function, which raises lysosomal pH (Kashyap et al., 2023), which further activates CTSH and leads to additional degradation of

Table 4
Druggability of proteins potentially causally associated with AD.

Target	Gene	Uniprot ID	Mediciation	Drug Introduction	Data source
Pro-cathepsin H	CTSH	P09668	p-Toluenesulfonic acid N-[1-Hydroxycarboxyethyl-Carbonyl]Leucylamino-2-Methyl-Butane	Experimental Experimental	Drugbank Drugbank
Triggering Receptor Expressed on Myeloid Cells 2	TREM2	Q9NZC2	Cyclocreatine	Cyclocreatine acts as a brain-permeable and effective bioenergetic protector by providing high levels of ATP, thereby reducing the atrophy of neurons adjacent to amyloid-β pathology in TREM2-deficient mice.	MCE
Progranulin	GRN	P28799	Latozinemab	Latozinemab effectively binds to Sortilin with high affinity and blocks the interaction between the progranulin (PGRN) and the Sortilin receptor. Latozinemab has the potential to cause the GRN mutation in the research of frontotemporal dementia (FTD-GRN).	MCE
Interleukin-34	IL34	Q6ZMJ4	AMG-820	The AMG-820 inhibits the binding of ligands CSF1 and IL34, as well as the subsequent activation of ligand-induced receptors.	MCE

functional GRN.

This study suggests that CTSH-GRN-TMEM106B and TREM2-IL-34 signaling networks may be involved in the pathological process of AD through interactions related to lysosomal dysfunction and neuro-inflammation. This study has identified several candidate drugs that target these pathways through drug screening based on the above-mentioned mechanism (Table 4). The first candidate is cyclocreatine, which acts as a brain-permeable bioenergetic protector. It has been revealed that rescuing neuronal atrophy near Aβ plaques in models deficient in TREM2 can be achieved by restoring ATP levels. The second candidate, Latozinemab, is a high-affinity anti-sortilin antibody that effectively blocks the degradation of progranulin, leading to an increase in extracellular GRN levels and helping mitigate lysosomal dysfunction, with demonstrated efficacy in studies related to FTD associated with GRN mutations. The third candidate, AMG-820, is an inhibitor of the CSF1R/IL-34 pathway, which prevents the activation of receptors induced by ligands and suppresses neuroinflammatory cascades. Furthermore, two drugs targeting CTSH were also identified, which are under development in DrugBank, and they exhibit promising results: p-Toluenesulfonic acid and N-[1-hydroxycarboxyethyl-carbonyl] leucylamino-2-methyl-butane. These novel compounds modulate CTSH activity and have the potential to regulate GRN processing within the CTSH-GRN-TMEM106B axis.

This study has several limitations. First, the current PPI analysis specifically focused on microglial networks. Future studies will extend interactome mapping to other critical cell types in the CNS, including astrocytes and endothelial cells, to comprehensively characterize cross-cellular regulatory mechanisms in AD pathogenesis. Second, while our neuropathological analyses of postmortem AD brain samples provide crucial human evidence for the spatial dysregulation of CTSH-GRN-TMEM106B and TREM2-IL-34 networks, these findings require validation through longitudinal intervention studies in animal models of AD. Additionally, during IHC and IF analyses, some cells in this study exhibited changes in cellular structure, as well as phenomena such as nuclear fragmentation and nuclear dissolution. These observations may be attributed to the constraints imposed by the time of death and the preservation conditions of the autopsy samples, which may have affected the results to some extent. Finally, the scRNA-seq data available for this study in public databases were sourced from the frontal cortex of brains of patients with AD from the in European populations, while the autopsy samples in this study were taken from the temporal cortex of AD brains in Asian populations. Further research is needed to provide more comprehensive foundational data.

5. Conclusions

This study used a comprehensive four-tiered verification framework in AD. This multi-omics joint analysis strategy revealed a microglia-centered interaction network characterized by coordinated lysosomal regulatory factors (CTSH-GRN-TMEM106B) and cytokine signaling components (IL-34-TREM2). This research identified potential avenues for developing therapies that target specific cell types. The findings of this research suggest the need to shift from traditional amyloid-focused approaches to a broader, integrative strategy that addresses the interactions between peripheral circulation and the CNS. This strategy may have profound implications for AD and be significant for additional tau-related diseases and disorders involving microglial participation.

CRediT authorship contribution statement

Yuhao Yuan: Writing – original draft. **Shengnan Wang:** Writing – original draft. **PiaoPiao Lian:** Methodology. **Zhonghao Yu:** Data curation. **Xiangting Gao:** Project administration. **Yijie Duan:** Methodology. **Ling Mao:** Writing – review & editing. **Yiwu Zhou:** Writing – review & editing.

Ethics declaration

Written informed consent was obtained from the legal representatives of the donors, and the research protocol was approved by the Ethics Committee of Huazhong University of Science and Technology. The relevant ethical materials are provided in the appendix, ethical number 2024 [S201].

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ejphar.2025.178297.

Appendix 1. Antibodies

Name	Source	Supplier	Article No.
SIGLEC-7	Rabbit	Proteintech	13939-1-AP
SIGLEC-9	Rabbit	Proteintech	13377-1-AP
CTSH	Rabbit	Proteintech	10315-1-AP
GRN	Rabbit	Proteintech	18410-1-AP
IL-34	Rabbit	Servicebio	GB111855
TMEM106B	Rabbit	Proteintech	20995-1-AP
TREM2	Rabbit	Proteintech	83438-6-RR
SIRPA	Rabbit	Servicebio	GB115426
Anti-Amyloid beta 40	Mouse	Servicebio	GB151197
Phospho-TAU	Rabbit	Servicebio	GB113883
Iba-1	Mouse	Santa	Sc-32725

Appendix 2. Abbreviation

(AD) Alzheimer's disease
(Aβ) Amyloid β-protein
(BBB) Blood-brain barrier
(CSF) Cerebrospinal fluid
(pQTLs) Protein quantitative trait loci
(MR) Mendelian randomization
(GWASs) Genome-wide association studies
(IVW) Inverse variance weighted
(FDR) False discovery rate
(snRNA-seq) Single-cell sequencing
(UMAP) Uniform manifold approximation and projection
(IHC) Immunohistochemistry
(CTSH) Cathepsin H
(SIRPA) Signal regulatory protein alpha
(TMEM106B) Transmembrane protein 106B
(BSP) Bone sialoprotein
(IL-34) Interleukin-34
(ILT-2) Immunoglobulin-like transcript 2
(GRN) Granulin
(TREM2) Triggering receptor expressed on myeloid cells 2
(SIGLEC9) Sialic acid-binding immunoglobulin-like lectin-9
(SNPs) Independent single nucleotide polymorphisms
(scRNA-seq) Single-cell RNA sequencing
(p-tau) Phosphorylated tau
(FTD) Frontotemporal dementia
(ALS) Amyotrophic lateral sclerosis

Data availability

All data produced or examined in this study can be found in the supplementary materials and data link.

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